

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 27

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Instability / landslips on the hillside, especially after heavy rain - compaction, drainage, establishing vegetation whose roots bind the spoil, regrading slopes • Acid drainage from oxidation of pyrites, with toxic metals - collect and treat leachate, add lime to raise pH • Spontaneous combustion of residual coal / hydrocarbons - layers of fine-grained material to reduce air flow • Lack of nutrients / poor plant growth - add topsoil, sewage sludge, fertiliser • Unattractive topography - landscaping / reshaping • Dust / heavy metals - cover and keep at high pH	6
01.2	• So that plant roots cannot reach the toxic layer • Otherwise toxins could be taken up and brought to the surface / dispersed via leaf litter and food chains	2
01.3	• Steel slag is alkaline, producing unusual calcareous / high-pH soil conditions • Supports uncommon plants such as cowslips / orchids that cannot compete on fertile neutral soils	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 282-283; AQA spec 3.4.3.2.12.	
02.1	• Igneous intrusions (batholiths) of magma cool within the crust, heating groundwater to produce superheated, pressurised water • The hot water dissolves minerals from the surrounding rock and the mineral-rich solutions travel along fissures / cracks away from the batholith • As the solutions cool the dissolved minerals crystallise in order of solubility (least soluble first) - fractional crystallisation • So different metals are separated and concentrated in veins, eg tin, copper, lead, silver, gold, arsenic (SW England batholith)	4

Question	Marking guidance	Marks
02.2	- Dissolved iron compounds in ancient seas were oxidised by the oxygen released by early photosynthetic organisms (2.5-1.8 billion years ago) - Iron oxides are insoluble so they were deposited as sediments (haematite, magnetite)	2
02.3	- Flowing water carries solid particles; the ability to carry them depends on the water velocity and the density of the solids - As velocity falls, dense minerals (gold, tin ore) are deposited first while lighter material is carried further, concentrating the heavy minerals in particular places	2
02.4	<i>Any 2 from:</i> - Limestone / chalk from the shells of marine organisms - Coal from terrestrial vegetation - Crude oil and natural gas from marine organisms (plankton)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 132-134; AQA spec 3.2.3.2.	
03.1	- Image databases: identifying individuals (tiger stripes, whale shark spots, dolphin fins) to find territory size, movements, lifespan, social groups - few species have unique recognisable features - eDNA: detecting the presence of a species from DNA shed into water (great crested newt, invasive carp) - shows presence not abundance; contamination / DNA transport by water - Geolocator tags: tracking small birds on migration where transmitters are too heavy - only record light and time so position is approximate; animal must be recaptured to download data	6
03.2	- Microwaves emitted by the satellite reflect off the surface giving data on wave height, wind velocity, sea surface altitude and oil slicks (which smooth the surface) - Microwaves pass through cloud so monitoring is not interrupted by weather	2
03.3	<i>Any 2 from:</i> - Droppings: diet, gender, territory size - Owl pellets: diet - Feeding marks (chewed hazel nuts - dormouse), tracks / footprints, nests / burrows, scratching posts (territory)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.7.2.3.	
04.1	- Single: $5.8 \times 20 \times 12 = 1392 \text{ W}$ - Argon: $1.6 \times 20 \times 12 = 384 \text{ W}$ - Reduction = 1008 W (accept 1010)	3
04.2	- A wider gap of trapped gas reduces conduction because gas is a poor conductor - If the gap is too wide convection currents start in the gas, carrying heat across; the optimum depends on the gas and window size	2
04.3	- Low-e coating reflects long-wavelength infrared back into the room, reducing radiation losses - Argon / krypton / xenon have lower thermal conductivity than air so less conduction across the gap	2

Question	Marking guidance	Marks
04.4	• Panes must be held apart by small spacers which cannot be long, so the gap is only about 0.2 mm • Used where there is not room for thick gas-filled multiple glazing (eg historic buildings)	2
04.5	• Each extra layer saves less energy (diminishing returns) while cost rises, so beyond the optimum the cost of insulation exceeds the value of energy saved	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-231; AQA spec 3.3.4.2.2.	
05.1	• Predicting the route / movement a pollutant will take through the environment (based on its properties and features of the environment) • To assess where it will become concentrated / which locations and organisms are most at risk, so monitoring can focus on those sites	2
05.2	<i>Any 3 from:</i> • Sea current velocity and direction • Properties of the radionuclides: half-life, solubility, tendency to adsorb onto sediments / bioaccumulate • Food chain pathways, eg uptake by seaweed, shellfish, fish eaten by people • Geology / hydrology of the coast; sediment deposition areas (mud flats) • Human activities in the area: fishing, seaweed collection, beach use	3
05.3	<i>Any 3 from:</i> • Identify the group of people whose lifestyle gives them the highest exposure (eg people who eat a lot of locally caught shellfish / fish, live near the mud flats, collect seaweed) • Monitor their exposure / the concentration of radioactive material in the foods they eat or their body • If their exposure / risk is acceptably low, everyone else is assumed to be at even lower risk; if not, emissions are controlled • CGM assesses exposure before health effects occur (not used to detect damage)	3
05.4	<i>Any 2 from:</i> • Seawater • Sediment / mud • Seaweed • Shellfish / molluscs / fish • Milk, vegetables or meat from nearby farms • Atmospheric dust • Soil	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243, 303-304; AQA spec 3.4.3.1.	

Question	Marking guidance	Marks
06.1	• Deforestation: less interception so less evaporation from leaves; less transpiration so less water returned to the atmosphere and less precipitation downwind; more infiltration and surface runoff, higher river flows / flooding • Urban development: impermeable concrete / tarmac reduce infiltration and aquifer recharge; runoff is faster and greater, increasing river flooding downstream; drains carry water quickly to rivers • Agriculture: soil compaction by machinery / trampling reduces infiltration and soil moisture, increases runoff; irrigation increases evaporation; loss of soil biota (worms) reduces infiltration and water retention; drainage lowers water tables	6
06.2	<i>Any 2 from:</i> • Higher temperatures increase evaporation and change the amount, timing, type (rain vs snow) and location of precipitation • More rapid melting of snow and ice changes river regimes: more flooding after rain, lower flows in dry periods • Changed wind patterns move humid air to different areas • Sea level rise / salt incursion into coastal aquifers	2
06.3	• Most water abstracted is cleaned and returned to the river / sea so the main impact is on quality (pollution) not quantity • Effects are noticeable only where large urban centres draw on restricted local supplies (aquifers, small rivers)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 116-117; AQA spec 3.2.2.1.	
07.1	• The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically)	1
07.2	<i>Any 4 from:</i> • Light: UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions • Temperature: higher temperatures increase the rate of chemical breakdown / microbial decomposition • Soil microbes / oxygen: aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils • pH / presence of other chemicals may increase or decrease the rate of degradation • Adsorption onto clay / organic matter may protect from degradation and increase persistence	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
08.1	• B	1
08.2	• TBT is highly toxic to marine molluscs / crustaceans and an endocrine disruptor altering growth and reproduction of oysters / whelks even at very low concentrations • It bioaccumulated in shellfish and could pass to humans eating them • It was assumed dilution would make it harmless (illustrates the need for the precautionary principle); now prohibited and replaced by copper	3

Question	Marking guidance	Marks
08.3	• In the low-oxygen conditions of the spoil heap iron is in its soluble reduced form and is leached into the river • In the river it is oxidised to insoluble iron oxide (orange sediment); the oxidation uses dissolved oxygen so the river is deoxygenated, killing aerobic organisms • Control: drainage water is passed over mesh screens where the iron oxidises and is deposited before the water reaches the river; the solid iron is removed periodically	3
08.4	<i>Any 2 from:</i> • Lead-acid car batteries (recycled) • Lead flashing on roofs / windows • Radiation shielding	2
08.5	• They inhibit enzyme function, especially the enzymes of the nervous system / nerve cells	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
09.1	• Temperature: most chemical / biological reactions are faster at higher temperature so degradable pollutants (eg sewage) break down faster; but faster decomposition may increase deoxygenation • Light: provides the activation energy for chemical reactions; photodegradation of pesticides (eg neonicotinoids break down fastest in sunlight); light drives photochemical smog reactions producing secondary pollutants • Oxygen: needed for aerobic bacterial decomposition of organic wastes / sewage; oxidation of sulfide ores produces sulfur dioxide / acid mine drainage; oxidation state affects the solubility of metals	6
09.2	<i>Any 2 from:</i> • Pollutant ions adsorb onto (the surface of) clay particles / organic matter in soil or sediment • So they are immobilised and cannot be absorbed by organisms / leached into water; radiation is absorbed in the ground • But they may be released later if conditions change (eg storms disturbing sediment, low pH)	2
09.3	• Many heavy metals (eg lead, aluminium) are more soluble at low pH • So they are mobilised / leached from soil or sediments and can be absorbed by organisms / enter water bodies	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 240-242, 255; AQA spec 3.4.2.2.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Properties of renewables: intermittency, predictability, low energy density, storage, applicability, locational constraints, impacts, technology level, economics • Wind: large offshore resource in shallow seas, falling costs, HAWT / VAWT / direct drive developments; intermittent, needs storage and grid; land-use conflicts • Solar: PV costs falling rapidly, roof space, but low winter insolation in the UK; CSP unsuitable; heat pumps and photothermal for heating • HEP: most large sites used; low-head, helical and micro-hydro extend use; pumped storage vital • Tidal: large range (Severn, Mersey), predictable; barrages costly with high impact, lagoons and in-stream turbines lower impact • Wave: long Atlantic fetch, but technology immature • Biofuels: wastes, anaerobic digestion, energy crops vs food / habitat • Geothermal: limited to low-temperature schemes / heat pumps • Systems issues: storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management, diversity of sources, energy conservation reducing demand, electrification of transport • Judgement: renewables could supply most electricity with storage and a diverse mix, but heat and transport are harder; transition needs investment and lifestyle change	9
10.2	• Eight times (energy proportional to velocity cubed)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-217; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
11.1	Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Hydrological cycle processes and reservoirs; residence times; dynamic equilibrium; increasing demand (population, affluence, irrigation, industry) • Human impacts on transfer rates: deforestation, agriculture, urbanisation, climate change, abstraction; consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • Case studies of unsustainable exploitation: Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • Sustainable strategies: metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair; new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis; pollution control (sewage, AMD, buffer strips) • Links to thermohaline circulation and climate; energy costs; circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	25

Question	Marking guidance	Marks
11.2	Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: toxicity (LD50, MDAF, MDNF), specificity, persistence, liposolubility/water solubility, mobility, mode of action (contact/systemic), synergism • Bioaccumulation and biomagnification explained by persistence + liposolubility; DDT and birds of prey as a case study • Comparison of organochlorines, organophosphates, pyrethroids, neonicotinoids with their relative advantages and disadvantages • Reducing impact: restrictions/bans, choice of non-persistent and specific pesticides, systemic pesticides and seed coatings, application timing (night, still days, not in flower), droplet size • Alternatives: cultural control (rotation, barrier/companion crops), biological control, predator habitats, sterile males, pheromone traps, genetic resistance, GM crops, integrated pest management • Trade-offs: yield, cost, human health (organophosphates), resistance evolution, pollinator services; risk:benefit; sustainability and the circular economy • Judgement: no pesticide is perfectly specific; combination of property-based selection and integrated control is most sustainable Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240, 262-266, 314-320; AQA spec 3.4.3.2.8, 3.5.1.1, 3.5.1.5.	25